WORKED SOLUTION

STEP II 2021, Question 2

Pure • Symmetric Functions of Roots

STEP II 2021, Question 2

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Sums of powers of two numbers are controlled by just two quantities: their sum and their product. Once you know $a+b$ and a^2+b^2, the product ab is forced, and so is a^3+b^3; that is the constraint in part (i). Part (iii) uses the same fact with c "removed": subtracting c, c^2, c^3 from the three-variable sums leaves two-variable sums, and the constraint from part (i) turns into a cubic equation that c must satisfy.

Solution

Part (i)

From $(a+b)^2 = a^2+2ab+b^2$ we get $p^2 = q+2ab$, so

$$ab = \frac{p^2-q}{2}.$$

Now use the factorisation $a^3+b^3 = (a+b)^3 - 3ab(a+b)$:

$$\begin{aligned} r = a^3+b^3 &= p^3 - 3p\cdot\frac{p^2-q}{2} \\ &= \frac{2p^3-3p^3+3pq}{2} = \frac{3pq-p^3}{2}. \end{aligned}$$

Hence $3pq-p^3 = 2r$, as required. Notice that nothing here used the fact that p, q, r are real: the identity holds for any complex a and b. We shall need that in part (iii).

Part (ii)

The quadratic $2x^2-2px+(p^2-q)=0$ always has two roots (possibly equal, possibly non-real), given by the quadratic formula. Call them a and b. Dividing through by 2, the quadratic is $x^2-px+\frac{1}{2}(p^2-q)$, so

$$a+b=p, \qquad ab=\frac{p^2-q}{2}.$$

This gives the first equation in $(*)$ directly. Then

$$a^2 + b^2 = (a+b)^2 - 2ab = p^2 - (p^2 - q) = q,$$

which is the second, and

$$a^3 + b^3 = (a+b)^3 - 3ab(a+b) = p^3 - \frac{3p(p^2-q)}{2} = \frac{3pq - p^3}{2} = r,$$

using the hypothesis $3pq - p^3 = 2r$ at the last step. So all three equations in $(*)$ hold.

The choice of quadratic was not luck: part (i) told us what ab has to be, and a quadratic with roots of known sum p and product $\frac{1}{2}(p^2 - q)$ is exactly $x^2 - px + \frac{1}{2}(p^2 - q)$.

Part (iii)

Set

$$p = a + b = s - c, \qquad q = a^2 + b^2 = t - c^2, \qquad r = a^3 + b^3 = u - c^3.$$

? **Ask yourself** Part (i) assumed p, q, r were real, but here they involve c, which may be complex. Can we still use it?

Yes: as noted at the end of part (i), the argument there is pure algebra and never used reality. So $3pq - p^3 = 2r$, that is,

$$3(s-c)(t-c^2) - (s-c)^3 = 2(u - c^3).$$

Expanding,

$$\begin{aligned} 3(s-c)(t-c^2) &= 3st - 3sc^2 - 3tc + 3c^3, \\ (s-c)^3 &= s^3 - 3s^2c + 3sc^2 - c^3, \end{aligned}$$

so the equation becomes

$$3st - 3sc^2 - 3tc + 3c^3 - s^3 + 3s^2c - 3sc^2 + c^3 = 2u - 2c^3.$$

Collecting powers of c:

$$6c^3 - 6sc^2 + 3(s^2 - t)c + 3st - s^3 - 2u = 0.$$

So c is a root of $6x^3 - 6sx^2 + 3(s^2 - t)x + 3st - s^3 - 2u = 0$, as required.

The coefficients depend only on s, t, u, which are symmetric in a, b, c. Repeating the argument with a (or b) in the role of c shows that a and b are roots too. So the other two roots are a and b.

? **Ask yourself** Why does it matter that a, b and c are distinct?

A cubic has at most three roots. Because a, b, c are three *different* roots, they are all of them, each simple, and so

$$6x^3 - 6sx^2 + 3(s^2 - t)x + 3st - s^3 - 2u = 6(x-a)(x-b)(x-c).$$

(If, say, $a = b$, we would only know two roots and could not identify the third.) Comparing constant terms, $3st - s^3 - 2u = -6abc = -6v$, that is,

$$s^3 - 3st + 2u = 6v.$$

Part (iv)

Here $s = 3$, $t = 1$, $u = -3$, $v = 2$. First a consistency check: the condition from part (iii) reads $27 - 9 - 6 = 12 = 6 \times 2$, so it holds. Part (iii) says that a, b, c should be the roots of

$$6x^3 - 18x^2 + 3(9-1)x + 9 - 27 + 6 = 6x^3 - 18x^2 + 24x - 12 = 0,$$

that is, of $x^3 - 3x^2 + 4x - 2 = 0$. Since $x = 1$ gives $1 - 3 + 4 - 2 = 0$,

$$x^3 - 3x^2 + 4x - 2 = (x-1)(x^2 - 2x + 2) = (x-1)((x-1)^2 + 1),$$

so the roots are 1, $1 + \mathrm{i}$, $1 - \mathrm{i}$. Take

$$a = 1, \qquad b = 1 + \mathrm{i}, \qquad c = 1 - \mathrm{i}.$$

Part (iii) only told us where to look, so we must now verify:

- $a + b + c = 1 + (1 + \mathrm{i}) + (1 - \mathrm{i}) = 3$;
- $(1 + \mathrm{i})^2 = 2\mathrm{i}$ and $(1 - \mathrm{i})^2 = -2\mathrm{i}$, so $a^2 + b^2 + c^2 = 1 + 2\mathrm{i} - 2\mathrm{i} = 1$;
- $(1 + \mathrm{i})^3 = 2\mathrm{i}(1 + \mathrm{i}) = -2 + 2\mathrm{i}$ and $(1 - \mathrm{i})^3 = -2\mathrm{i}(1 - \mathrm{i}) = -2 - 2\mathrm{i}$, so $a^3 + b^3 + c^3 = 1 - 4 = -3$;
- $abc = 1 \cdot (1 + \mathrm{i})(1 - \mathrm{i}) = 1 \cdot 2 = 2$.

All four equations $(**)$ hold. (Any rearrangement of these three numbers works equally well.)

Checks

- Part (i) with $a = 1$, $b = 2$: $p = 3$, $q = 5$, $r = 9$, and $3pq - p^3 = 45 - 27 = 18 = 2r$.
- Part (iii) with $a, b, c = 1, 2, 3$: $s = 6$, $t = 14$, $u = 36$, $v = 6$, and $s^3 - 3st + 2u = 216 - 252 + 72 = 36 = 6v$. The cubic is $6x^3 - 36x^2 + 66x - 36 = 6(x-1)(x-2)(x-3)$, as it should be.
- The identities in parts (i)–(iii) were also confirmed by computer for many random complex a, b, c.

Common slips

- In part (iii), saying "a and b are the other roots" without noting that the cubic's coefficients are symmetric in a, b, c.

- Deducing $s^3 - 3st + 2u = 6v$ from the product of roots without using distinctness to know that a, b, c are *all* the roots.
- In part (iv), stopping at the roots of the cubic: the question asks you to verify the four equations explicitly.

Beyond STEP

This question is a small case of Newton's identities, which link the power sums $a^k + b^k + c^k$ to the elementary symmetric functions $a + b + c$, $ab + bc + ca$, abc. The fundamental theorem of symmetric polynomials says that every symmetric polynomial can be written in terms of the elementary ones, and this is the starting point of Galois theory. For more, see Allenby, *Rings, Fields and Groups*.

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